

EDIConnect

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Chapter 1

Introduction

EDI, which stands for Electronic Data Interchange, is the inter-company communication of business documents in a standard format. It has been widely viewed as a technology pivotal to supply chain management that has also provided benefits to firms on multiple levels. It replaces paper-based documents such as purchase orders or invoices with online documents. These EDI files usually follow a standard, which contains segments, elements, and other contextual features allowing users to add their data accurately into the specific sections.

Internationally, there are multiple standards being followed in different industries, for example, Tradacoms, X12, EDIFACT, EANCOM, SWIFT, HL7, etc. In addition, sometimes some companies created their own custom standards for localised use. In B2B (business-to-business) transactions, these EDI standards are quite helpful.

Furthermore, companies mostly use ERP systems (Enterprise Resource Planning) to manage and integrate the important parts of their businesses, to connect planning, purchasing inventory, sales, marketing, finance, human resources, and more. So during the transactions, the information being sent and received by the users is stored in the ERP systems.

Therefore, we can say that the information contained by the EDIs is fed into the ERP systems, and through this process, a company deals with their customers efficiently. Globally, there are various ERP systems being used, like Microsoft Dynamics 365, SAP (Systems Applications and Products), etc.

Hundreds and thousands of transactions are being done by the companies in a single day. Having a seamless mechanism to handle the transactional data will be beneficial for the companies. Internationally, there are more than 300 standards being used across various industries. Therefore, having an interpreter of messages being exchanged by the companies will play a crucial role. Therefore, generating mappings can allow users to have a better knowledge of the data.

Moreover, if we look into the process of how these EDIs help customers, we'll be able to

understand that the crucial role after getting the EDI is to interpret the information and to enter it into their own ERP systems. A slight mistake in this process may lead to a huge financial setback for the companies.

Since the EDI data of each file is in a diverse structure and format according to the standards they are written in. Whether it be EDIFACT, X12, or TRADACOMS, etc., they organise information in their own unique way. This makes the process of aligning or converting data fields from one format to another relatively difficult. The task of interpreting and finding the corresponding fields within these standards requires manual effort, especially when dealing with a high volume of data or multiple standards.

Additionally, mapping EDI data to the ERP system's format faces complexities. ERP systems have their own specific data structures that may not align directly with accurate mapping between EDI documents and ERP fields, which makes things more critical to ensure that data flows seamlessly into the ERP system without errors. Misalignment in these mappings can result in incorrect data entry, processing errors, and business disruptions.

Therefore, there is an overriding need to generate mappings to comprehend the message. Ensuring that all information is accurately captured and processed is vital to maintaining the smooth operation of business activities. Users are required to depict the important information from the EDI file to store it into their ERP system. Hence, the need for a mapping tool becomes essential for the users to apprehend and map the data being exchanged into their own standard and their own used ERP system.

To evaluate our project, we consider multiple standards and custom EDI files as case studies. Our aim is to create a seamless system generating the mappings of EDI files from one standard to another and furthermore providing users mappings against the ERP system entered by the user. The mappings being done in our system will be customisable, allowing users to get a better understanding of the information being exchanged.[2]

1.1 Problem Statement

In the business world, Electronic Data Interchange (EDI) plays a major role in inter-company communication, but the diversity of EDI standards and custom formats creates significant challenges. Information contained by the EDI files is fed into the Enterprise Resource Planning (ERP) systems, and through this process, a company deals with their customers efficiently. Globally, there are various ERP systems being used, like Microsoft 365, SAP, etc. Companies are required to convert the data into their own used EDI standard and map that data into the ERP systems.

These mappings take a lot of time and human energy; therefore, our project addresses these issues by developing a system that not only translates EDI data from one interna-

tionally declared standard or custom standards to another but also creates customisable mappings for EDI data from one standard to another standard and into specific ERP system formats. This way, it will allow businesses to address their specific needs, enhancing flexibility and effectiveness in handling diverse data requirements.

1.2 Scope

Our goal is to reduce hindrances that occurred in business due to differences in EDI standards and ERP systems. It makes it convenient for the companies to apprehend the message and generate a mapping that will make it easy for the user to enter the message into their respective ERP systems.

Our tool, which is a customisable front-end, will provide the user with the ability to easily comprehend the information being exchanged and will generate mappings from one standard to another standard of EDI file, custom EDI files to standard EDI using complex semantic mapping techniques and rule-based NLP techniques where necessary. Additionally, we will also provide mappings for EDI to ERP systems in acceptable format, but our tool won't provide any sort of integration facility. Following are the objectives of our project:

- Identify EDI standard or custom EDI file.
- Data pre-processing of EDI files sent by user.
- Generate mappings across standards.
- Generate mappings from custom EDI file EDI standard.
- Generate mappings for EDI file to user's ERP system accepted format.
- User friendly interface

1.3 Modules

1.3.1 Module 1

In this module, we will be generating rule-based mappings from one internationally declared EDI standard to another. We will be providing users the facility to create EDI standard to standard mapping templates, allowing them to effortlessly map data in less time.

1. Dataset creation in MongoDB.
2. Define rules.
3. Contextualise an open source LLM and implement RAG.
4. Apply NLP techniques.
5. Template creation of most commonly used transaction set.
6. Testing and validation.

1.4 User Classes and Characteristics

User class	Description
Business Analytics	Business analytics are the employees who manage these business documents of their company on daily basis. They need a seamless system to map the data into their used standard, which can be an internationally declared one or a custom one, on daily basis. Regular use of the system can allow them to generate, update and, administer the data mappings effortless. Additionally, validation between the data being exchanged is a critical component of business dealings, ensuring accuracy and compliance in every transaction
IT Administrators	IT administrators are the people who are responsible for maintaining, configuring, and troubleshooting the system. There is a constant urge to check the data received is accurate, properly mapped, and stored in ERP systems being used by the company.
Logistics Managers	As Logistics managers are responsible for movement, distribution and storage of data. They play a major role in coordinating with suppliers, transportation providers, and internal teams to ensure smooth operations ensuring that the accurate data exchanged between different systems and to validate it essentially just to avoid disruptions in the supply chain. There is a major need to ensure that the EDI documents are timely exchanged, and with highest accuracy to manage supply chain operations efficiently.
Finance Department Employees	Finance department's major roles evolve around making sure that the exchange of transactional data is accurately done between the company partners. The system will allow users to reduce the risk of errors that could lead to payment delays etc.
Warehouse Managers	Incoming and Outgoing data exchange is a major task for a warehouse. Therefore, Warehouse managers are required to precisely describe the information being exchanged.
End Users	These are the non-technical employees who interact using EDI documents. Reviewing and processing the exchanged documents is done by them on daily basis that's why they constantly need an interface that allows them to easily interact with the EDI system without needing technical knowledge of how the mapping works.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case Diagrams

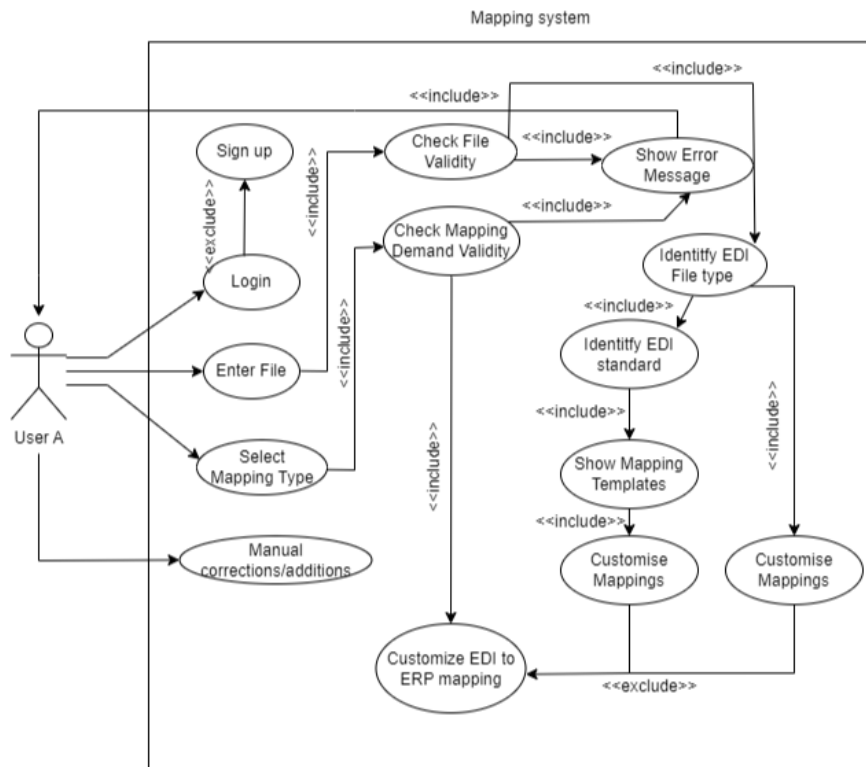


Figure 2.1: Use case Diagram

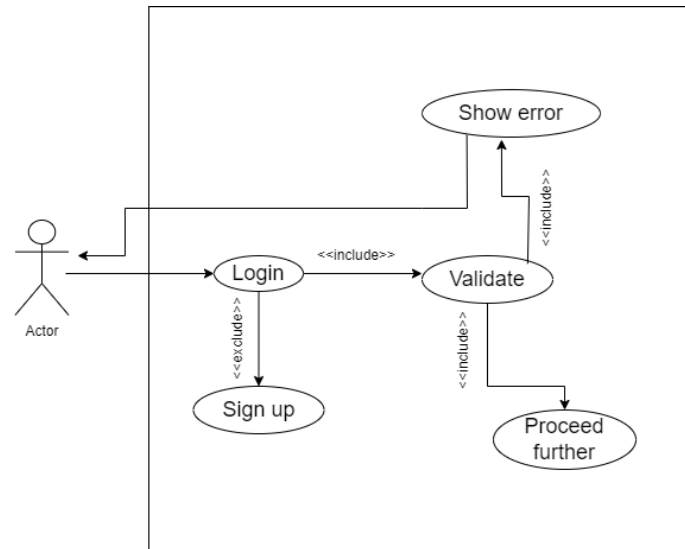


Figure 2.2: Login/SignUp use case

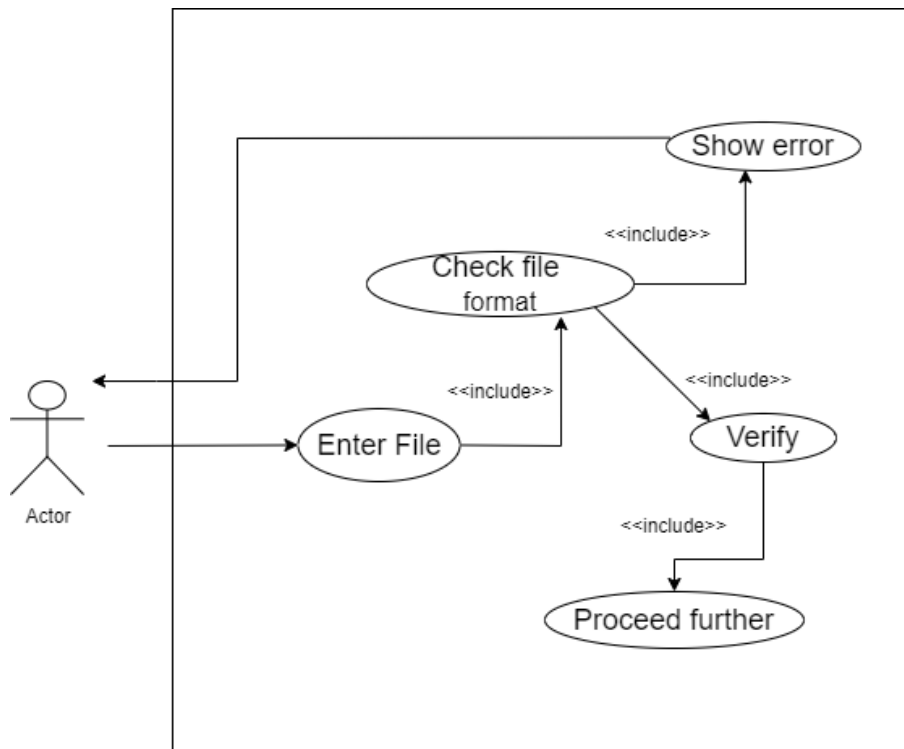


Figure 2.3: File Format Validity use case

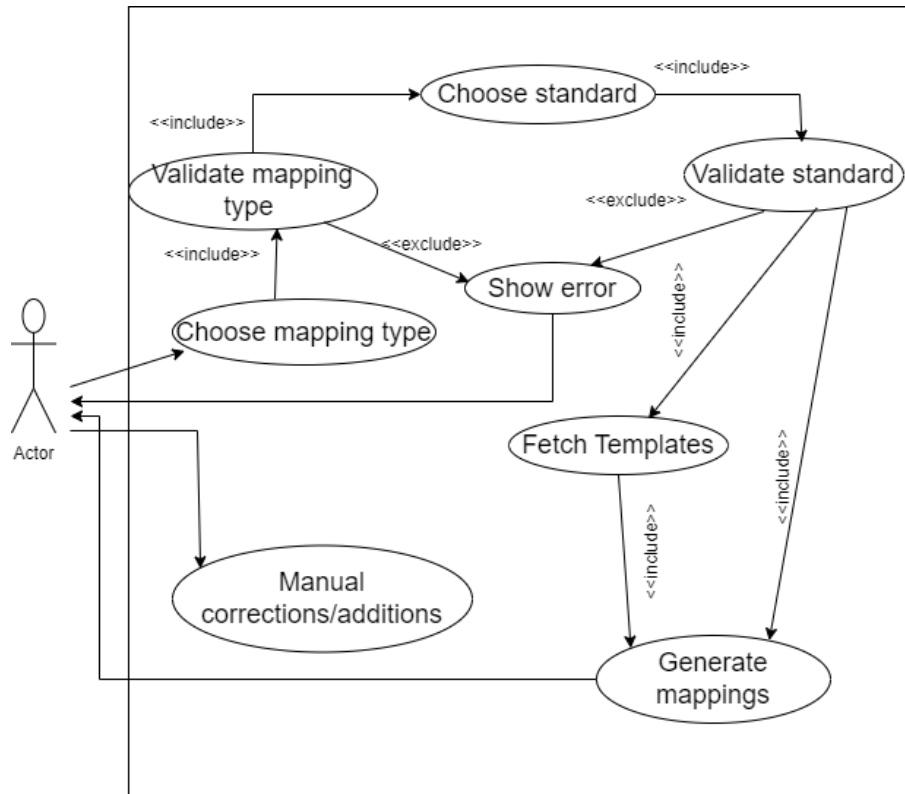


Figure 2.4: Mapping requirement and template usecase diagram

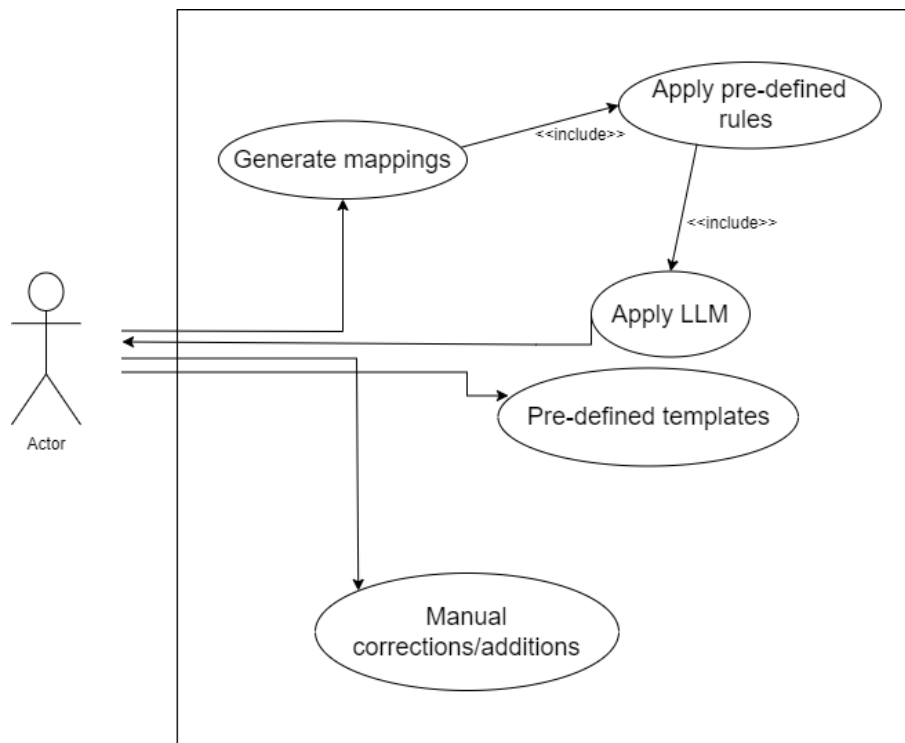


Figure 2.5: EDI to EDI use case

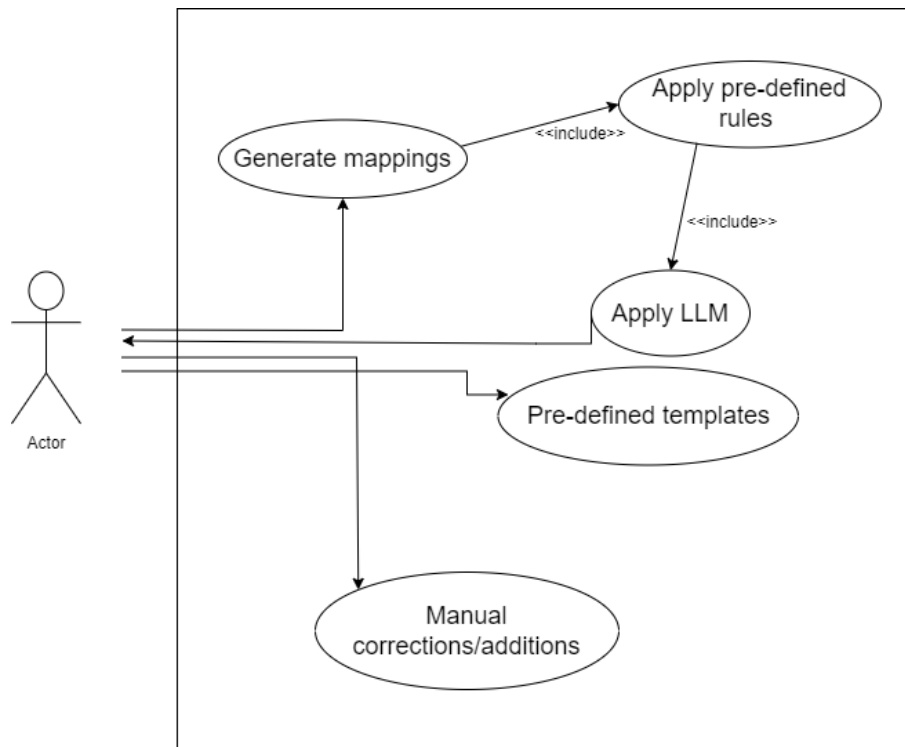


Figure 2.6: EDI to ERP use case

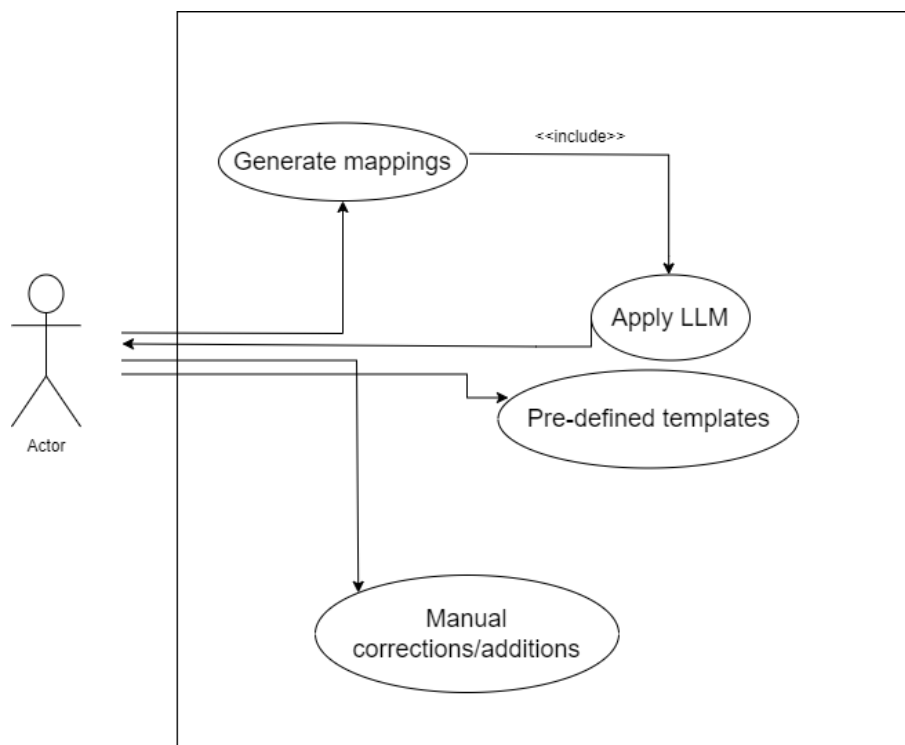


Figure 2.7: Non-EDI to EDI use case

2.2 Functional Requirements

Our system will provide a UI that will allow users to input their EDI file to generate mappings between different EDI standards. It will allow users to efficiently map EDI data from one standard to another, reducing time and effort. The system will allow users to input an EDI file in the respective standard they use and choose the standard in which they want to convert. The system will automatically detect the standard and pre-process the data and will start mapping from the input standard to the standard required by the user.

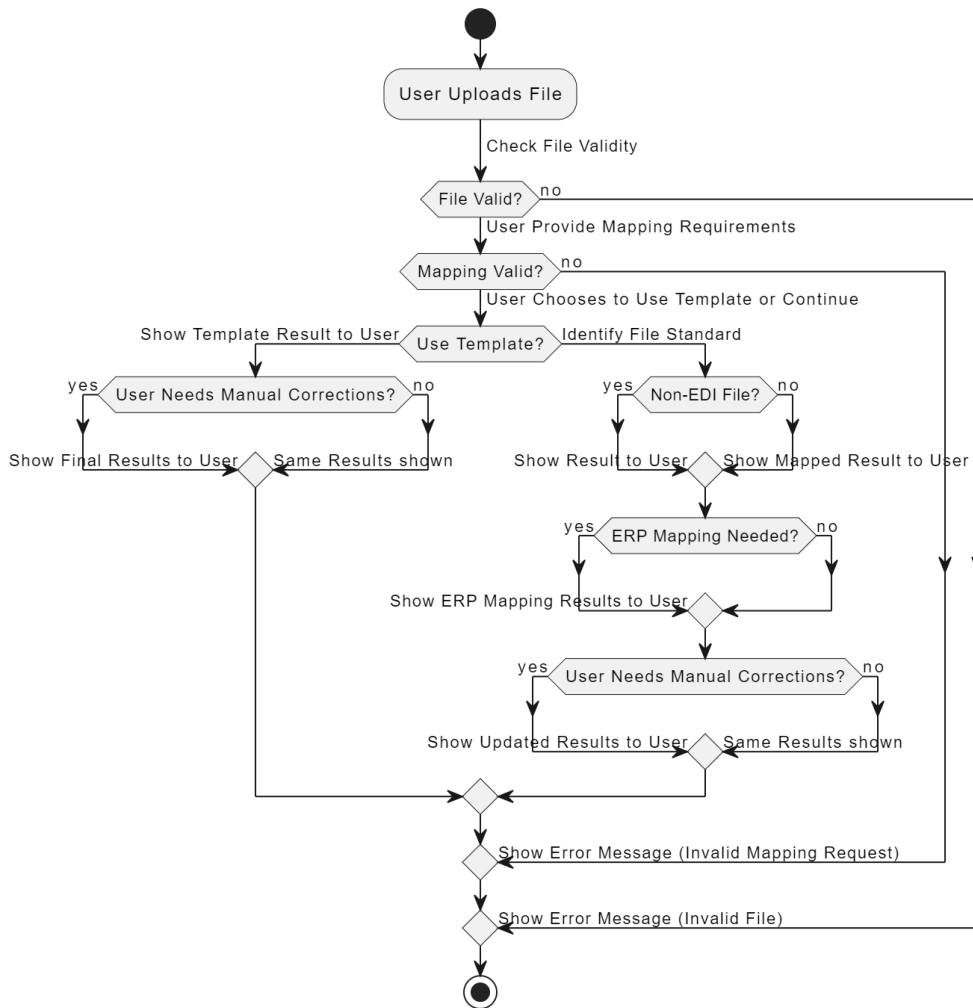


Figure 2.8: Process Flow Diagram for Functional Requirements

2.2.1 Module 1

Following are the requirements for Module 1:

1. UI that will allow users to input their EDI file to generate mappings between different EDI standards.
2. The UI will ask the user to enter their mapping demand, stating which kind of mapping they need (EDI to EDI, non-EDI to EDI, or EDI to ERP), and will also ask for the standards for converting files from and to.
3. If the EDI file belongs to the standards covered in our scope, the system will also give the user some examples of standard-to-standard mappings based on predefined templates.

2.3 Non-Functional Requirements

2.3.1 Reliability

The system shall generate clear error messages if the system encounters any issue while uploading EDI files or requesting a mapping not supported by the system. Identifying file type and confirming it's supported by our system, and making sure the file is uploaded by the user correctly, will allow us to monitor any issue that arises during this process. Our system shall allow the user to upload the file and firstly choose the option of EDI to EDI mapping or EDI to ERP mapping, then it shall be able to check the EDI file to see if it is in the format that our system supports. It is a necessity for the EDI file to belong to the list of specific standards of either EDI or ERP that we follow and to lie within the industries specified in our scope. However, in the end, once the system shows an error message to the user, the user will be asked to upload their mapping demands or the accurate file. Additionally, we'll allow users to manually correct or to add data into the generated final mappings.

2.3.2 Usability

The system shall provide an intuitive and user-friendly interface for input and visualisation by the user. The system initially will ask users to enter the type of mapping they need along with the format to convert the EDI file being entered. Then asks the user to enter the file and gives them the option to opt for templates or generate mapping using LLMs, etc. By showing the file entered by the user side by side to the mapping result generated by the system, it will allow them to effortlessly comprehend how the data is being mapped from one format to another format. Enabling users to do manual additions for any missing information they require will enable user to produce more accurate results.

2.3.3 Security

The system shall secure user credentials for safe login and sign up.

2.4 Domain Model

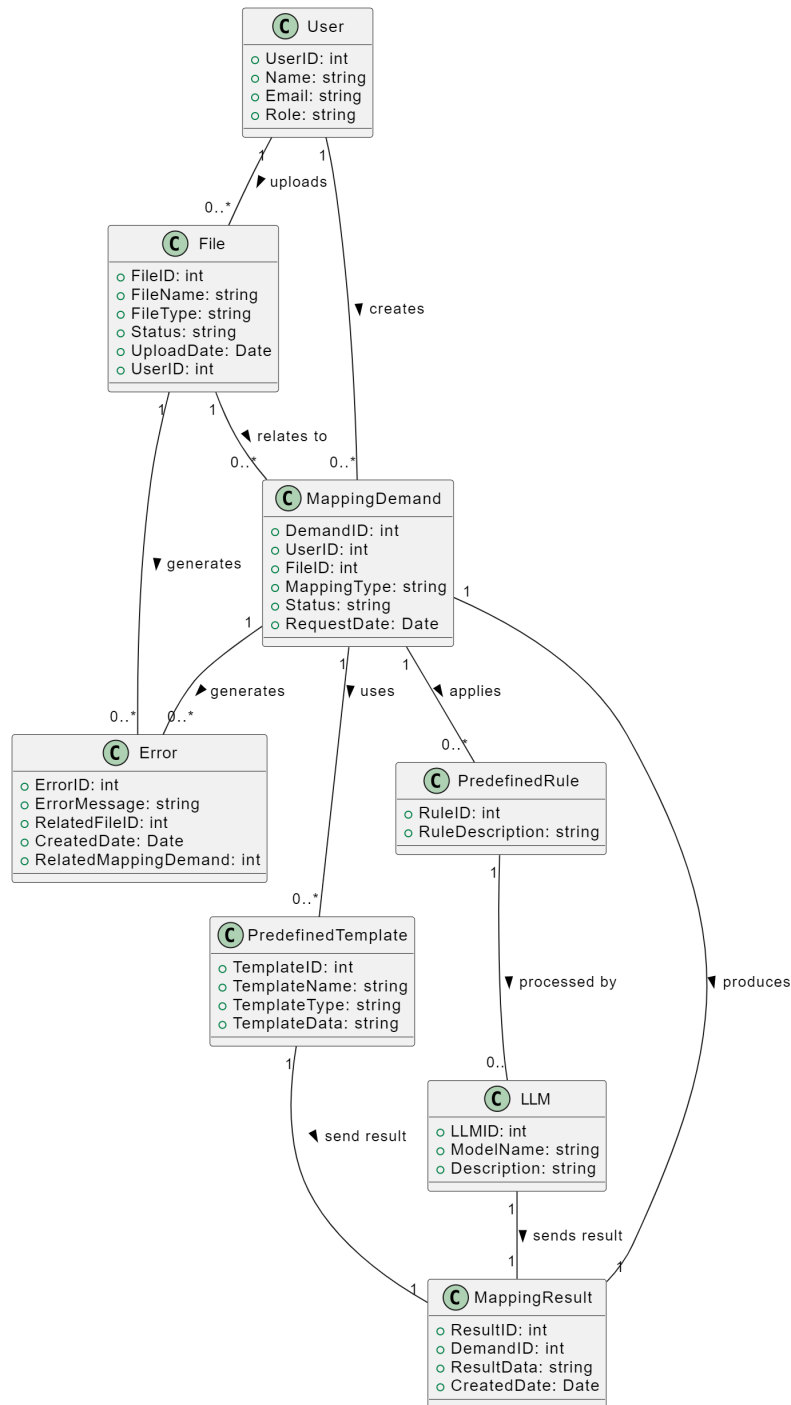


Figure 2.9: Domain model.

Chapter 3

System Overview

3.1 Architectural Design

With the high complexity of EDI standards and ERP formats having an architecture that differentiates between each of the layers. Layered Architecture will clearly define how mapping, LLM integration, validation, and user management are done all interact in structured layers, while keeping the design easy to understand and manage. Since there is no such need for having independent scaling, therefore, there's no such need left for microservices architecture. A layered architecture enables

1. Maintainability
2. Scalability
3. Reliability

The architecture is defined into 3 layers as mentioned bellow:

1. Presentation Layer:
 - (a) UI module: This module allows users to interact with the system. It will allow end users to upload their EDI documents, specify mapping requirements, perform validations, and view the generated mappings. Additionally, users can opt to use predefined templates for quicker and more efficient mapping processes, providing a seamless and intuitive experience.
2. Business Logic Layer:
 - (a) Mapping engine: This is the core module of the system we are building, as it generates the mappings using the pre-defined set of rules for all sorts of mappings. This module interacts with the LLM module for further enhancements in the mappings generated.

- (a) LLM integration: An open-source module will be used along with RAGs to enhance the mappings generated by the pre-defined rules, adapting them for more complex or customized requirements.

3. Data Layer:

- (a) Data storage: Here all sorts of EDI and ERP standards our system supports and their structure is stored. Pre-defined templates for the user's ease are stored along with the EDI files uploaded by the user into the system.

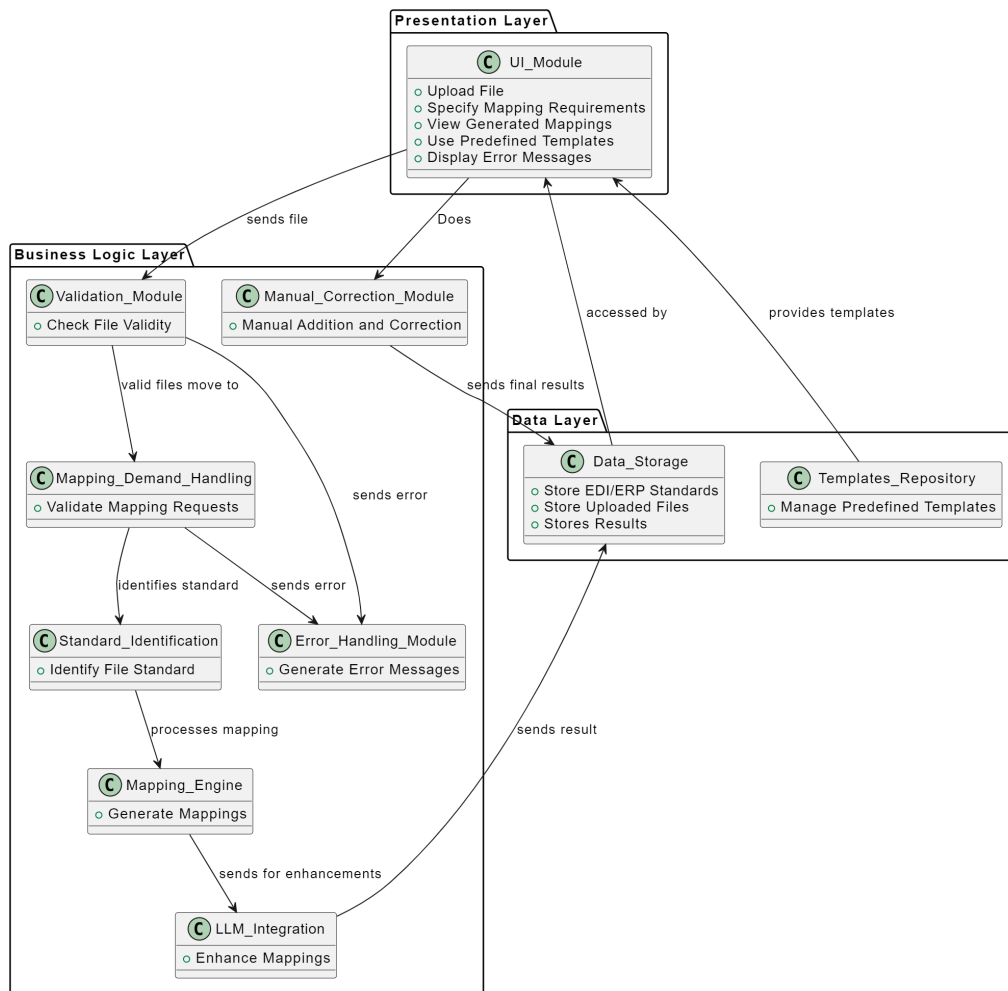


Figure 3.1: Layered Architecture for System overview.

3.1.1 System Sequence Diagram

The sequence diagram of EDIConnect is the interaction diagram that details how operations are carried out. It captures the interaction between objects in the context of a

collaboration. It shows the order of the interaction visually by using the vertical axis of the diagram to represent at what time what sort of messages are sent to which process.

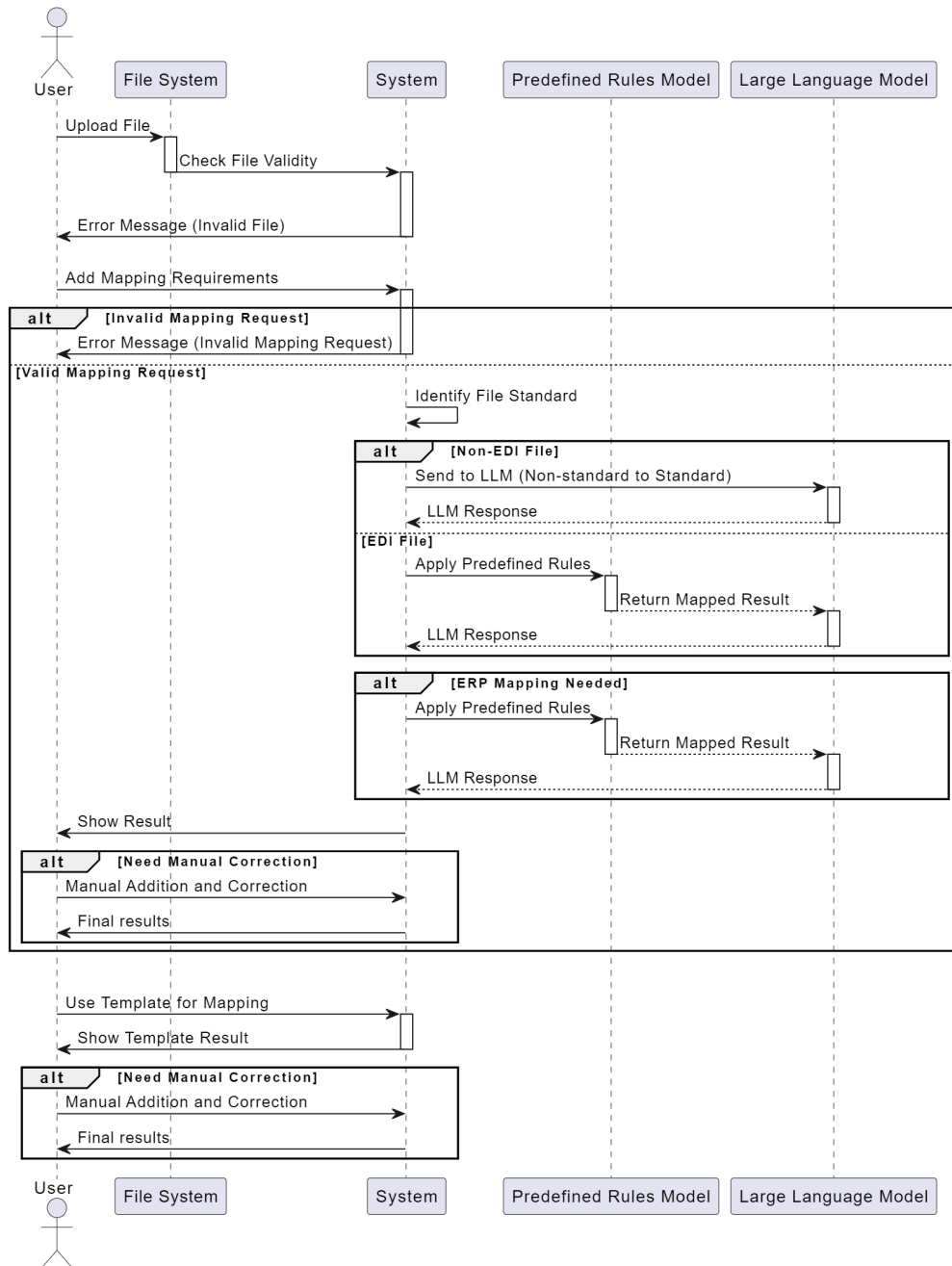


Figure 3.2: System Sequence Diagram.

3.1.2 Data Flow Diagrams

1. Data Flow Diagram Level 0

This level of the project will show you a top-level, simplified version of the data flow in EDIConnect. It will be catering to the initial interaction of the user with the UI system. The user will be able to input the respective file, either a standard or a custom file. The user will then receive an acknowledgement or error message in case the file has any sort of errors or missing content, and then after the acknowledgement, the user will get the required mapped data.

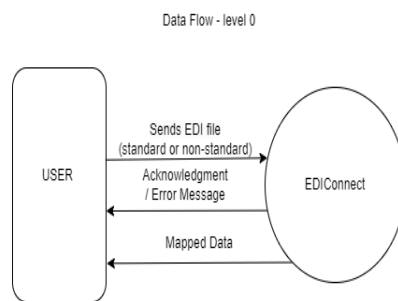


Figure 3.3: Data Flow Diagram Level 0.

2. Data Flow Diagram Level 1

Level 1 delves deeper into the sub-processes of EDIConnect, specifically showing the handling of EDI file transfers and processing. This level explains how the standard is identified and sent to the data store for rule-based mapping, leading to output as required by the user.

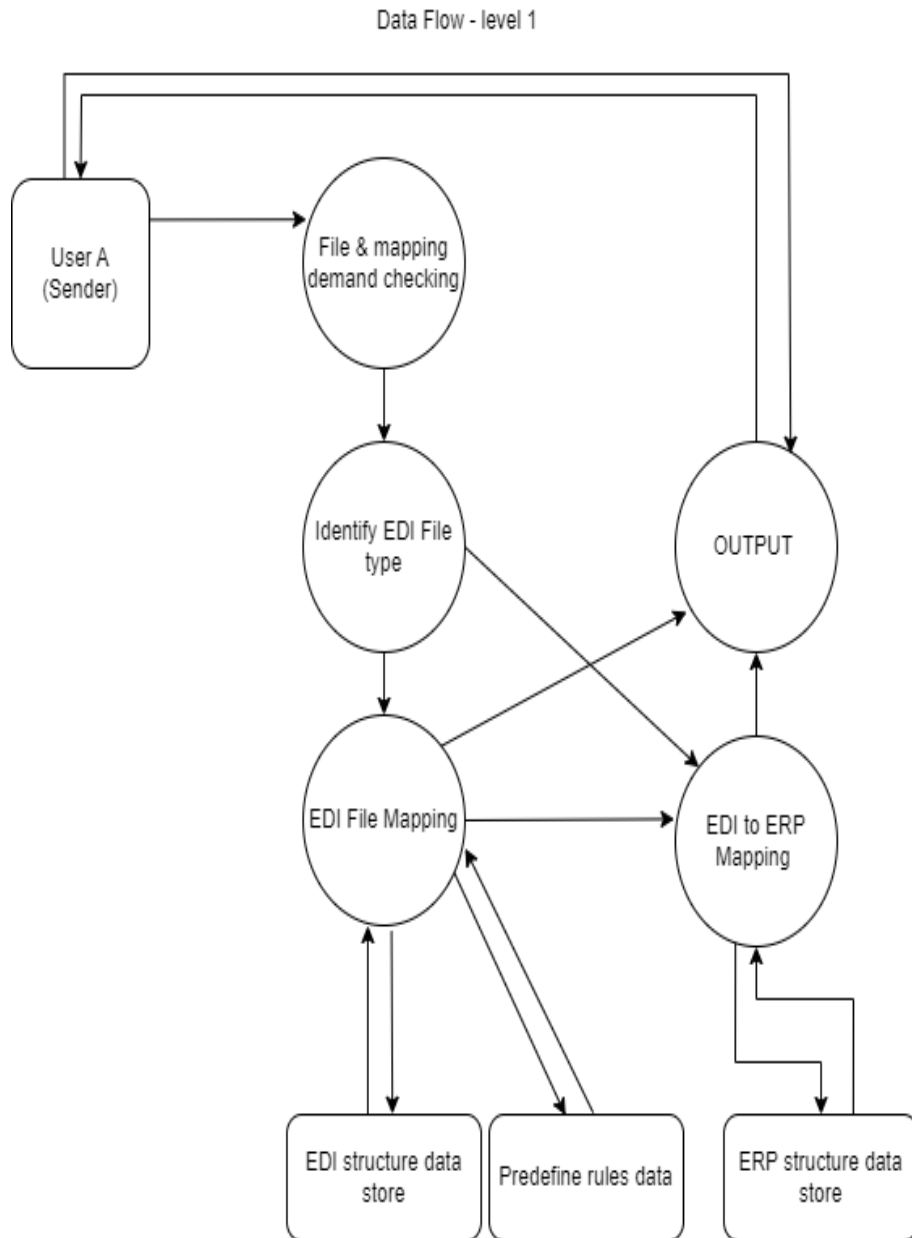


Figure 3.4: Data Flow Diagram Level 1.

3. Data Flow Diagram Level 2

This level provides more granular information about EDIConnect's processes, including where NLP techniques and LLMs validate and map data, and how the file flows from user input to desired output.

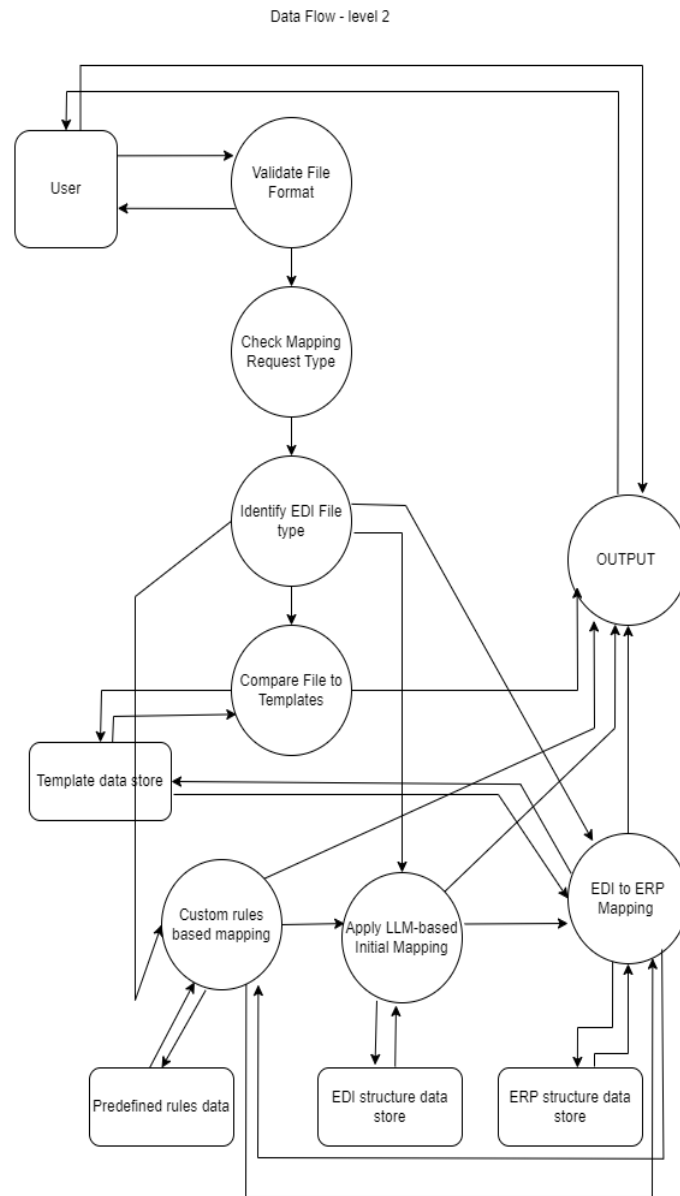


Figure 3.5: Data Flow Diagram Level 2.

4. Data Flow Diagram Level 3

This level provides an in-depth view of the project's architecture, detailing the linkage between the front-end, back-end, and dataset. It illustrates the continuous flow of EDI files, showcasing how files are converted between standards with proper mapping of mandatory fields.

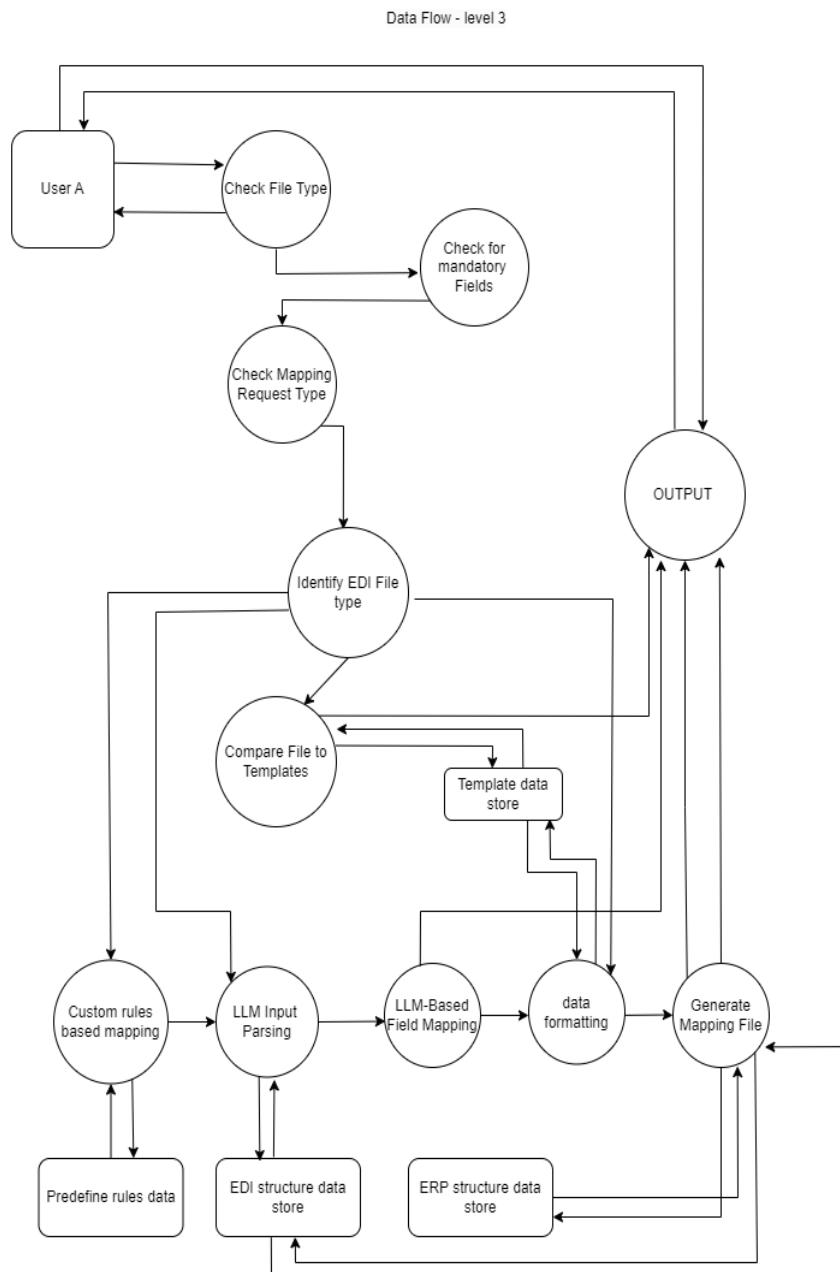


Figure 3.6: Data Flow Diagram Level 3.

3.1.3 State Transmission Diagram

This state transmission diagram shows the process flow of how files are handled and mapped across the whole pipeline within EDIConnect . The file after uploading is checked and validated, it is an EDI or non-EDI format, predefined rules or an LLM are applied for processing. The flow continues with either automated or manual corrections, leading to the final mapped output. If errors occur, they are handled appropriately during the process.

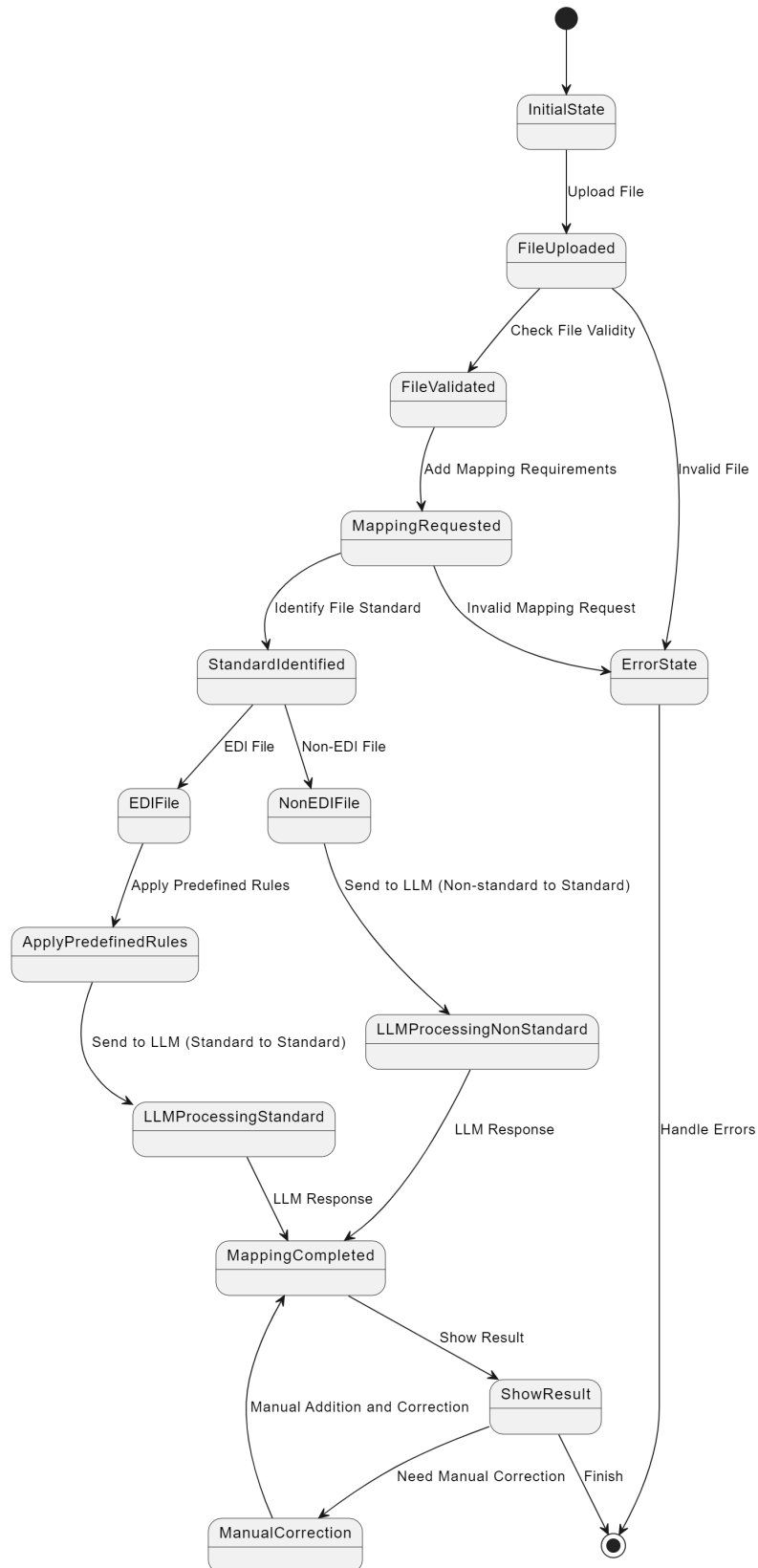


Figure 3.7: State Transmission Diagram.

3.1.4 System Architecture Diagram

The architecture diagram outlines the overall structure of the EDIConnect system. It begins with the validation of file structure and identification of the file standard, enabling standard-to-standard conversions using predefined rules. The system also utilizes LLMs for contextual mapping, particularly when dealing with non-standard formats or complex scenarios such as custom EDI to standard EDI mappings. In cases where ERP integration is required, the system leverages NLP similarity techniques for accurate mapping and uses the LLM for further optimization. The user interacts with this system via an intuitive interface, where they can input a file, select a target format or standard, and review the converted output.

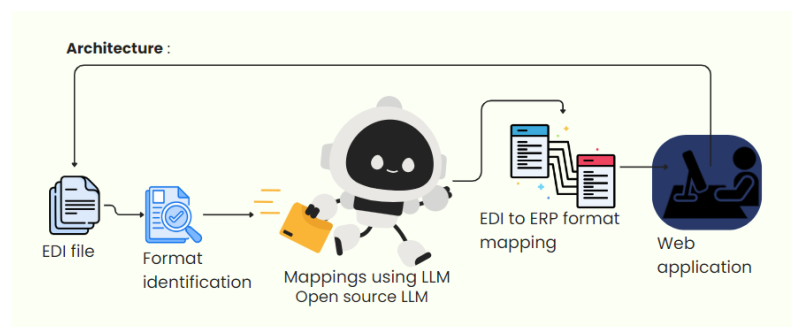


Figure 3.8: System Architecture Diagram.

3.2 Data Design

3.2.1 Information Domain

The information domain precisely defines how and where data is collected. Since our project lacked an explicit dataset, the data was gathered from authentic sources such as *stedi* and *HL7* websites [3] [1], which provided the necessary data standards and rules. The collection includes relevant data, concepts, rules, and constraints that the system must adhere to.

In our case, for an EDI application, the information domain consists of:

- **EDI Standards:** Various EDI formats like EDIFACT, X12, and their corresponding fields, segments (or elements), transaction sets, and along with this EDI files are also stored here.
- **Business Rules:** Rules that govern how data should be transformed or validated between systems (e.g., mapping of invoice data between different formats).
- **Mapping Templates:** Rule-based or NLP-generated templates that define the mapping between different EDI standards.
- **ERP Formats:** Various EDI formats like Microsoft Dynamics 365, SAP, and their corresponding fields and files.
- **Custom EDI files:** Diverse non-EDI files and their corresponding fields, without any specific format followed.

The structure of the data hierarchy is represented as follows:

[Standard Name] → [Transaction Set Info] → [Segments] → [Elements]

The following is an example of how this structure might be stored:

```
[ {"X12 8050"},
  {"transaction_set_number": "103",
   "transaction_set_name": "Abandoned Property Filings"},
  [ {"transaction_set_number": "103",
     "position": "0100", "segment": "ST",
     "name": "Transaction Set Header", "category": "Mandatory"}
  ],
  [ {"element_name": "Route Code", "element_number": "1",
```

```

    "description": "Mutually defined route code", "length": "Min"},
    {"element_name": "Adjustment Number", "element_number": "31",
      "description": "Identification assigned to an inventory level"
      , "length": "Min"},
    {"element_name": "Equipment Description Code",
      "element_number": "40",
      "description": "Code identifying type of equipment used",
      "length": "Min"},
    {"element_name": "Delivery Date", "element_number": "3
2",
      "description": "Delivery date format CCYYMMDD", "length": "Min"}
  ]
]

```

3.2.2 Data Structures Transformation

Data structure transformation refers to the process of converting data from one format or structure to another. This is crucial in EDI systems, as different standards or versions have varying structures. Some types of data transformation include:

- **Syntax Transformation:** Converting data between various syntax (e.g., XML to JSON, or X12 to EDIFACT).
- **Schema Mapping:** Mapping fields from one schema to another, ensuring that the data conforms to the target system's requirements (e.g., mapping fields from an X12 850 Purchase Order to an EDIFACT ORDERS transaction).
- **Element/Segment Mapping:** Transforming individual elements or segments within a transaction set from one structure to another based on predefined or dynamic rules (e.g., mapping "Invoice Amount" from a custom format to a standard format).
- **Data Validation and Enrichment:** Validating the data during transformation and possibly enriching it with additional information (e.g., adding missing fields or converting units).

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